

8 Statistics

8.3 Describing and Summarizing Data

Let us start by finding a description of the “middle” of a dataset.

Definition 8.3.1 The **mean** of a set of data is the sum the data values divided by the number of values.

Example 8.3.2 Marci’s exam scores for her last math class were: 79, 86, 82, 94. Find the mean of her exam scores.

Example 8.3.3 The number of touchdown (TD) passes thrown by each of the 31 teams in the National Football League in the 2000 season are shown below.

The total is 634, average is 20.45.

$\{37, 33, 33, 32, 29, 28, 28, 23, 22, 22, 22, 21, 21, 21, 20, 20, 19, 19, 18, 18, 18, 18, 16, 15, 14, 14, 14, 12, 12, 9, 6\}$.

Example 8.3.4 The one hundred families in a particular neighborhood are asked their annual household income, to the nearest \$5 thousand dollars. The results are summarized in a frequency table below.

Income	Frequency
15	6
20	8
25	11
30	17
35	19
40	20
45	12
50	7

Find the mean annual income for the neighborhood.

Use the fact that we have the frequencies to compute a sort of “weighted mean”, which is 33.9

Example 8.3.5 Suppose that a new family moves into the neighborhood with an income of \$5 million, what is the new mean?

Definition 8.3.6 An **outlier** is a data point which lies “far” from the other data points (we will give a better definition of this later)

When our data contains outliers, it is often better to measure the “center” of our data using the median:

Definition 8.3.7 The **median** of a set of data is the value in the middle of the *ordered* data.

Example 8.3.8 Find the median of the following data set:

$$\{5, 3, 7, 9, 2, 5, 10, 11, 4, 2, 6, 6, 1\}$$

Write out the formulas for which data point we need:

$$\frac{N + 1}{2}$$

notice that when N is even, this means we need to take an average of two data points.

Example 8.3.9 Let’s revisit the income example, which contained the outlier.

Income	Frequency
15	6
20	8
25	11
30	17
35	19
40	20
45	12
50	7
5000	1

Find the median annual income for the neighborhood.

Example 8.3.10 Find the median of the following data set:

$$\{3, 1, 5, 8, 3, 2, 2, 5, 4, 0, 1, 1, 4\}.$$

We will next describe how “spread out” the data is.

Definition 8.3.11 The standard deviation is a measure of how far data points are from the mean, it has a few important qualities:

1. It is always positive, and is only zero if all the data is the same.
2. It has the same units as the original data.
3. It can be highly influenced by outliers, just like the mean.

Definition 8.3.12 To compute the standard deviation:

1. Calculate the mean of the data.
2. Compute the difference of each data point and the mean.
3. Square the differences.
4. Compute the mean of the squares*.
5. Take a square root

*Unless the data represents a population, we divide by $N - 1$ instead of N .

Example 8.3.13 Marci’s exam scores for her last math class were: 79, 86, 82, 94. Find the mean of her exam scores, then find the standard deviation.

Example 8.3.14 The price of a jar of peanut butter at 5 stores were: \$3.29, \$3.59, \$3.79, \$3.75, and \$3.99. Find the standard deviation of the prices.

Definition 8.3.15 An **outlier** is a data point which lies more than 2 standard deviations away from the mean.

Example 8.3.16 Let’s revisit the income example.

Income	Frequency
15	6
20	8
25	11
30	17
35	19
40	20
45	12
50	7
5000	1

Identify any outliers in this data.